SHRUTHI CHANNARAM

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PROFESSIONAL SUMMARY

Highly motivated Electronics & Communication Engineer with an M.Tech in Wireless and Mobile Communication and a strong academic background from Diploma to Post-Graduation. Hands-on experience in PCB design, RF testing, communication systems, and hardware validation through industrial Training and Apprenticeship. Strong foundation in Digital & Analog Communication, MIMO-OFDM, Signal Processing, Embedded Systems, VLSI, and IoT, with basic knowledge of Python, C, and MATLAB for simulation and algorithm development. Seeking opportunities in RF, Communication, and Embedded domains.

PROFESSIONAL EXPERIENCE

- **Apprenticeship – Bharat Electronics Limited (BEL)**

Duration: JANUARY 2024 – DECEMBER 2025

  - Completed 1-year apprenticeship during M.Tech in the ECE domain, focusing on communication systems, hardware testing, and system validation.
  - Gained hands-on experience in signal processing, embedded system integration.
  - Assisted in the Development & Engineering (D&E) Department, in EWLS (Electronics Warfare Land System) working on Direction Finding, transmission and reception modules in communication systems, as well as jamming techniques.

PROFESSIONAL DEVELOPMENT

- **Industrial Training (6 Months) PCB Design Trainee – National Small Industries Corporation (NSIC)**

Duration: 2020

  - Undertook 6 - month industrial training during Diploma in PCB Design including schematic capture, multilayer layout, and circuit testing.
  - Worked with design tools and testing instruments to develop and validate PCB prototypes.
  - Strengthened skills in electronics design, PC Hardware and documentation.

KEY SKILLS

- Good foundation in Electronics, Digital & Analog Communication systems, modulation techniques, and wireless communication principles.
- Academic understanding of Deep Learning concepts, including Convolutional Autoencoder (CAE), along with core electronics fundamentals.
- Theoretical knowledge of Arduino, IoT architecture, circuit simulation, and basic PCB design concepts.
- Familiarity with electronic test instruments such as Spectrum Analyzer, Signal Generator, and Multi-meter.
- Basic knowledge in Python programming, supported by strong problem-solving and technical documentation skills.

Publications

- Published research paper titled *“A Comparative Study of Traditional PAPR Reduction Techniques in MIMO-OFDM Systems”* in the International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering (IJIREEICE).
- Published research paper titled *“IoT-Based Cardiac Arrest Detection System”* in the International Journal for Research in Applied Science & Engineering Technology (IJRASET).

EDUCATION

- **M.Tech in Wireless and Mobile Communication(WMC)**
  G. Narayanamma Institute of Technology and Science, Hyderabad, India 2023-2025
- **B.Tech in Electronics and Communication Engineering (ECE)**
  KG Reddy College of Engineering and Technology, Hyderabad, India 2020-2023
- **Diploma in Electronics and Communication Engineering (ECE)**
  Bandari Srinivas Institute of Technology, Hyderabad 2017-2020
- **Secondary Education (Class X)**
  TS Model School, SSC 2017

PROJECTS

1. **PAPR Reduction in MIMO-OFDM through CAE Based Optimization (M.Tech Major Project)**
   - Designed and simulated MIMO-OFDM systems in MATLAB for wireless communication.
   - Implemented PAPR reduction techniques such as PTS, SLM, and Convolutional Autoencoder (CAE).
   - Evaluated system performance using CCDF, BER, and SER analysis under different channel models.

2. **IoT-Based Advanced Cardiac Arrest Detection System (M.Tech Mini Project)**
   - Designed a system for early detection of cardiac arrest conditions.
   - Incorporated automated emergency alerts with real-time patient status transmission.

3. **IoT-Based Accident Detection and Rescue System (B.Tech Major Project)**
   - Built a system for real-time accident detection using IoT sensors.
   - Implemented an automated rescue alert mechanism with GPS-based location tracking.

4. **IoT Based Patients Health Monitoring System (B.Tech Mini Project)**
   - Developed an IoT-based system for real-time monitoring of patient health parameters.
   - Integrated wireless data transmission with an automated alert mechanism.

5. **Sensor-Based Smart Wheelchair using Arduino Uno (Diploma Project)**
   - Developed a sensor-controlled wheelchair for mobility assistance.
   - Implemented obstacle detection and control system to enhance safety and convenience.

6. **Accident Detection System using GPS, GSM, Arduino, and Accelerometer Sensor (Diploma Project)**
   - Designed a low-cost accident detection prototype using Arduino and sensors.
   - Enabled automatic location tracking and emergency alert transmission via GSM/GPS.

PERSONAL INFORMATION

Name: SHRUTHI CHANNARAM
DOB: 4th MARCH 2001
Father: BALASWAMY CHANNARAM
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